

seminar 1

PROGRAMMING LANGUAGE SPECIFICATION

1) BNF (Backus-Naur Form)

Language elements:

- meta linguistic variables (nonterminal) - written between $\langle \rangle$
- language primitives (terminals) - written between delimiters
- meta linguistic connectors
 - $::=$ (equals by definition)
 - $|$ (or)

example 1. all nonempty sequences of letters.

$\langle \text{lettsq} \rangle ::= \langle \text{letter} \rangle | \langle \text{letter} \rangle \langle \text{lettsq} \rangle$

$\langle \text{letter} \rangle ::= a | b | \dots | z | A | B | \dots | Z$

example 2. all signed and unsigned integers with the following constraints:

- 0 is unsigned

- numbers of at least 2 digits should not start with 0

$\langle \text{nonzerodigit} \rangle ::= 1 | 2 | \dots | 9$

$\langle \text{digit} \rangle ::= 0 | \langle \text{nonzerodigit} \rangle$

$\langle \text{sign} \rangle ::= - | +$

$\langle \text{integer} \rangle ::= 0 | \langle \text{unsignedint} \rangle | \langle \text{signedint} \rangle$

$\langle \text{digitseq} \rangle ::= \langle \text{digit} \rangle | \langle \text{digit} \rangle \langle \text{digitseq} \rangle$

$\langle \text{unsignedint} \rangle ::= \langle \text{nonzerodigit} \rangle |$
 $\langle \text{nonzerodigit} \rangle \langle \text{digitseq} \rangle$

$\langle \text{signedint} \rangle ::= \langle \text{sign} \rangle \langle \text{unsignedint} \rangle$

2) EBNF (Extended BNF)

Wirth's dialect

- nonterminals without $\langle \rangle$
- terminals between " "
- $::=$ becomes $=$
- $\{ \}$ - 0 or more
- $[]$ - optionality (0 or 1)
- $()$ - for grouping
- $(* *)$ - comments
- rules end with $.$

example 3, rewrite example 2 in EBNF.

integer = "0" | ["-" | "+"] nonzerodigit { digit }

nonzerodigit = "1" | "2" | ... | "9"

digit = "0" | nonzero

seminar 2

SCANNING ALGORITHM

$E = E + "T" | E - "T" | T$

$T = T * "F" | T / "F" | F$

$F = "(" E ")" | id | NOCONST$

example 1. Program.txt

program test;

var a, b: integer;

c: string;

begin

a := 1;

if (a >= b) then

b := 2;

write("message");

end

input: text file containing the program list of tokens from "tokens.txt"

output: PIF + ST + lexical errors (if any)

PIF

ST - only id + consts

token	st_pos	st_pos	symbol
program	-1	0	test
id	0	1	a
;	-1	2	b
var	-1	3	c
id	1	4	1
,	-1	5	2
id	2	6	"message"
:	-1		
integer	-1		
;	-1		
id	3		
:	-1		
string	-1		
;	-1		
begin	-1		
id	1		
:=	-1		
const	4		
;	-1		
if	-1		
(	-1		

seminar 3

GRAMMARS

$$G = (N, \Sigma, P, S)$$

P - set of production

$$P \subseteq (N \cup \Sigma)^* N (N \cup \Sigma)^* X (N \cup \Sigma)^*$$

$$N \cup \Sigma = V$$

$$(N \cup \Sigma)^* = V^*$$

ϵ - empty sequence
(epsilon, not sigma)

$$\alpha \rightarrow \beta \Leftrightarrow (\alpha, \beta)$$

$$\alpha \Rightarrow \beta \text{ iff } \alpha = \alpha_1 \gamma_1 \beta_1 \text{ \& } \beta = \alpha_1 \gamma_2 \beta_1 \text{ \& } \gamma_1 \rightarrow \gamma_2 \in P$$

$$\overset{n}{\Rightarrow} n \in \mathbb{N}^*$$

$$\overset{*}{\Rightarrow}$$

$$\overset{+}{\Rightarrow}$$

direct derivation

$$L(G) = \{ w \in \Sigma^* \mid S^* \rightarrow w \} - \text{language generated by the grammar}$$

ex. 1 $G = (N, \Sigma, P, S)$

$$N = \{S, C\}, \Sigma = \{a, b\}$$

$$P: S \rightarrow ab \mid aCSb$$

$$C \rightarrow S \mid bSb$$

$$CS \rightarrow b$$

$$? w = ab(ab^2)^2 \in L(G)$$

$$(1) S \rightarrow$$

$$(2) S \rightarrow$$

$$(3) C \rightarrow$$

$$(4) C \rightarrow$$

$$(5) CS \rightarrow$$

$$(ab)^2 \neq$$

$$abab$$

$$S \Rightarrow_2 a$$

ex. 2 $G = (N, \Sigma, P, S)$

$$N = \{S\}, \Sigma = \{a, b\}$$

$$P: S \rightarrow a$$

$$? L(G)$$

$$L = \{ a^{2^n} b \}$$

$$? L = L(G)$$

$$1 ? L \subseteq L(G)$$

$$2 ? L(G) \subseteq L$$

$$1 ? \forall w \in L$$

$$\forall n \in \mathbb{N},$$

$$A. n=0 \Rightarrow$$

B. Assume

Prove P1

$$① S \rightarrow ab$$

$$② S \rightarrow aCSb$$

$$③ C \rightarrow S$$

$$④ C \rightarrow bSb$$

$$⑤ CS \rightarrow b$$

$$(ab)^2 \neq a^2b^2$$

$$\begin{matrix} \text{"} \\ abab \end{matrix} \quad \begin{matrix} \text{"} \\ aabb \end{matrix}$$

$$S \xRightarrow{2} aCSb \xRightarrow{4} abSbSb \xRightarrow{2,1} ab(ab^2)^2 \Rightarrow S \xRightarrow{4} w \Rightarrow w \in L(G)$$

ex. 2 $G = (N, \Sigma, P, S)$

$$N = \{S\}, \Sigma = \{a, b, c\}$$

$$P: S \rightarrow a^2S | bc$$

$$? L(G)$$

$$L = \{a^{2n}bc \mid n \in \mathbb{N}\}$$

$$? L = L(G)$$

$$1? L \subseteq L(G)$$

$$2? L(G) \subseteq L$$

$$1? \forall w \in L, w \in L(G)$$

$\forall n \in \mathbb{N}, a^{2n}bc \in L(G)$ (induction) $P(n): a^{2n}bc \in L(G), n \in \mathbb{N}$ to prove that $P(n)$ true, $\forall n \in \mathbb{N}$

$$A. n=0 \Rightarrow bc \in L(G) \text{ True}$$

$$S \xRightarrow{2} bc$$

$$B. \text{ Assume } P(n): a^{2n}bc \in L(G) \text{ is True} \Rightarrow S \xRightarrow{n} a^{2n}bc$$

Prove $P(n+1)$ is True

$$P(n+1): a^{2n+2}bc = a^2 \cdot a^{2n}bc$$

$$S \Rightarrow a^2 S \xRightarrow[\text{induction}]{*} a^2 \cdot a^{2n}bc \Rightarrow P(n+1) \text{ True}$$

From $A+B \Rightarrow P(n)$ is True, $\forall n \in \mathbb{N}$

$$2? L(G) \subseteq L$$

$$S \Rightarrow bc = a^{2 \cdot 0}bc \in L$$

$$\Rightarrow a^2 S \Rightarrow a^2 bc \in L$$

$$\Rightarrow a^4 S \Rightarrow a^4 bc \in L$$

$$\Rightarrow \dots$$

! true only if you used all combinations in production

$$\text{ex. 3 } L = \{0^n 1^n 2^m \mid n, m \in \mathbb{N}^*\}$$

$$? G \text{ s.t. } L = L(G)$$

$$S \Rightarrow AB$$

$$A \rightarrow 0A1 \mid 01$$

$$B \rightarrow 2 \mid 2B$$

$$1. L(G) \subseteq L$$

$$2. L \subseteq L(G)$$

$$1. S \Rightarrow AB$$

$$A \Rightarrow 01$$

$$\Rightarrow 0A1 \Rightarrow 0^2 1^2$$

$$\Rightarrow 0^2 A 1^2 \Rightarrow 0^3 1^3$$

$$\Rightarrow 0^3 A 1^3 \Rightarrow \dots$$

$$\Rightarrow \dots$$

(i) A only generates seq. of the shape $0^n 1^n, n \in \mathbb{N}^*$

$$B \Rightarrow 2$$

$$\Rightarrow 2B \Rightarrow 2^2$$

$$\Rightarrow 2^2 B \Rightarrow 2^3$$

$$\Rightarrow 2^3 B \Rightarrow \dots$$

$$\Rightarrow \dots$$

ii) B only generates seq. of terms 5

$$2^m, m \in \mathbb{N}^*$$

From (i), (ii), $\Rightarrow S$ only generates seq. of terms of the shape $0^n 1^n 2^m, n, m \in \mathbb{N}^*$

2. $\forall n, m \in \mathbb{N}^* \Rightarrow 0^n 1^n 2^m \in L(G)$

Let n, m be fixed ($\in \mathbb{N}^*$)

$$S \Rightarrow AB \xRightarrow[(i)]{n} 0^n 1^n B \xRightarrow[(ii)]{m} 0^n 1^n 2^m \Rightarrow S \xRightarrow{1+n+m} 0^n 1^n 2^m \Rightarrow 0^n 1^n 2^m \in L(G)$$

$$(i) A \xRightarrow{n} 0^n 1^n, \forall n \in \mathbb{N}^*$$

$$(ii) B \xRightarrow{m} 2^m, \forall m \in \mathbb{N}^*$$

(i): $P(k): A \xRightarrow{k} 0^k 1^k, k \in \mathbb{N}^*$ and prove $P(k)$ true $\forall k \in \mathbb{N}^*$ by math ind.

A. $k=1 \Rightarrow A \Rightarrow 0_1$ (true based on 3)

B. $P(k) \rightarrow P(k+1)$

$[P(k) \Rightarrow A \xRightarrow{k} 0^k 1^k]$ - ind. hyp.

$$P(k+1): A \xRightarrow{k+1} 0^{k+1} 1^{k+1}$$

$$A \xRightarrow{2} 0A1 \xRightarrow[\text{hyp.}]{k} 00^k 1^k 1 = 0^{k+1} 1^{k+1} \Rightarrow A \xRightarrow{k+1} 0^{k+1} 1^{k+1} \Rightarrow P(k+1) \text{ true}$$

$\Rightarrow$ proof step true

seminar 4

FINITE AUTOMATA

$$FA: M = \{Q, \Sigma, \delta, q_0, F\}$$

$$Q = \{q_0, q_1, q_2, q_3, q_7\}$$

$$\Sigma = \{1, 2, 3\}$$

$$F = \{q_7\}$$

exercise 1.

δ	1	2	3	
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$	$\{q_0, q_3\}$	0
q_1	$\{q_1, q_2\}$	$\{q_1\}$	$\{q_1\}$	0
q_2	$\{q_2\}$	$\{q_2, q_7\}$	$\{q_2\}$	0
q_3	$\{q_3\}$	$\{q_3\}$	$\{q_3, q_7\}$	0
q_7	$\emptyset$	$\emptyset$	$\emptyset$	1

$$\delta: Q \times \Sigma \rightarrow P(Q)$$

$$(q, x) \text{ config.}$$

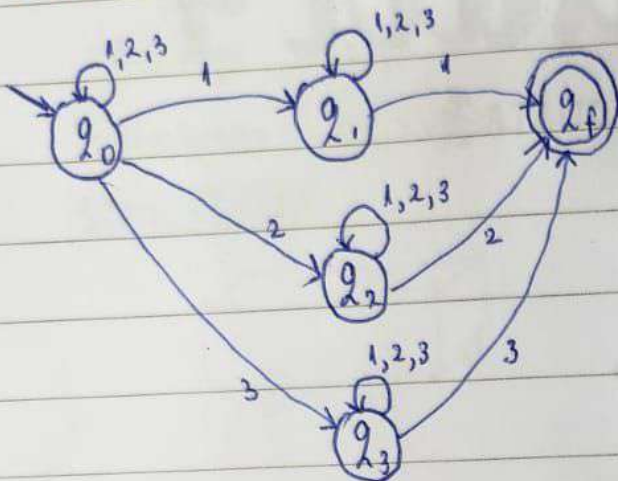
$$(q_0, w) \text{ - initial config.}$$

$$(q_7, \Sigma) \text{ - final config.}$$

$$(P, ax) \vdash (q, x) \text{ iff } q \in \delta(P, a)$$

$$L(M) = \{w \in \Sigma^* \mid (q_0, w) \vdash^* (q_7, \Sigma), q \in \delta\}$$

? $w = 12321 \in L(M)$

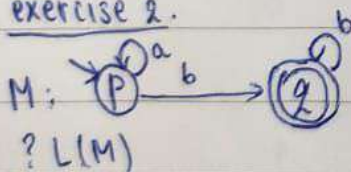


$(q_0, 12321) \vdash (q_1, 2321) \vdash (q_1, 321) \vdash (q_1, 21) \vdash (q_1, 1) \vdash$
 $\vdash (q_f, \Sigma)$

$\Rightarrow (q_0, w) \vdash^5 (q_f, \Sigma)$

$w \in L(M)$

exercise 2.



$L = \{a^n b^m, n \in \mathbb{N}, m \in \mathbb{N}^*\}$

? 1) $L \subseteq L(M)$

? 2) $L(M) \subseteq L$

1) if $n \in \mathbb{N}, m \in \mathbb{N}^*, a^n b^m \in L(M)$

Let $n \in \mathbb{N}, m \in \mathbb{N}^*$ be fixed.

$(p, a^n b^m) \vdash^n_a (p, b^m) \vdash (q, b^{m-1}) \vdash^{m-1}_b (q, \Sigma)$

a) $(p, a^n) \vdash^n (p, \Sigma), n \in \mathbb{N}$

b) $(q, b^m) \vdash^m (q, \Sigma), m \in \mathbb{N}$

$$a) P(n): (p, a^n) \vdash^n (p, \Sigma), n \in \mathbb{N}$$

A. verification

$$n=0 \quad (p, \Sigma) \vdash^0 (p, \Sigma) \text{ True}$$

B. proof step

$$P(k) \rightarrow P(k+1), k \in \mathbb{N}$$

$$P(k): (p, a^k) \vdash^k (p, \Sigma), k \in \mathbb{N}$$

$$(p, a^{k+1}) \vdash (p, a^k) \stackrel{(\text{ind. hyp.})}{\vdash^k} (p, \Sigma)$$

$$\Rightarrow P(k+1) \text{ true}$$

$$(p, a^n b^m) \vdash^{n+m} (q, \Sigma) \Rightarrow a^n b^m \in L(M)$$

$$2) \quad a^n \cdot b \cdot b^k, n, k \in \mathbb{N}$$

$$a^n \cdot b^m, n \in \mathbb{N}, m \in \mathbb{N}^*$$

ALL possible paths from initial state to the ^{all} final states.

exercise 3. ? FA

$$a) L = \{0^n 1^m 2^k \mid n, m \in \mathbb{N}, k \in \mathbb{N}\}$$

$$b) L = \{0(01)^n \mid n \in \mathbb{N}\}$$

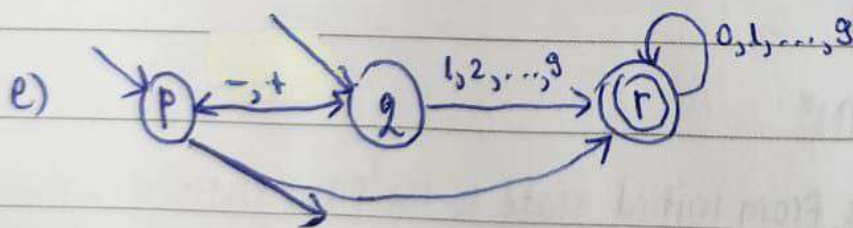
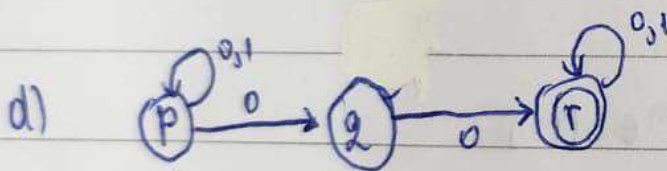
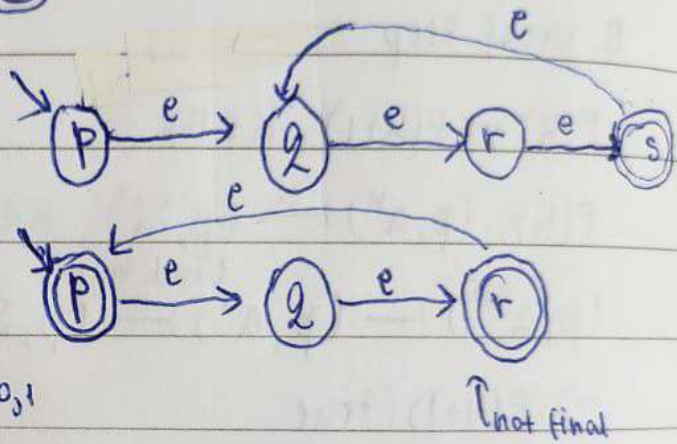
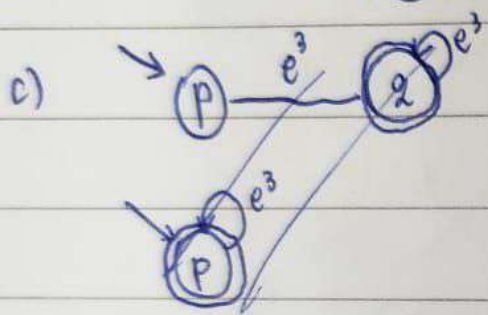
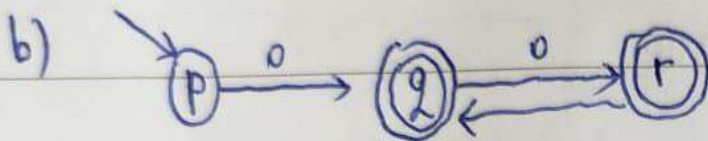
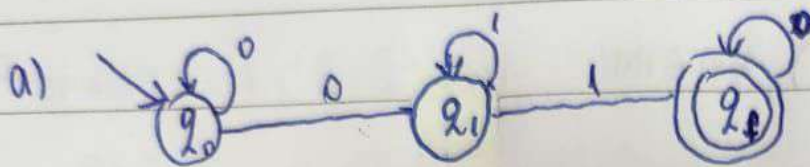
$$c) L = \{e^{3^n} \mid n \in \mathbb{N}\}$$

$$L = \{e^{3^n} \mid n \in \mathbb{N}\}$$

$$d) \Sigma = \{0, 1\} + \text{all sequences have at least two consecutive 0}$$

e) integers

f)



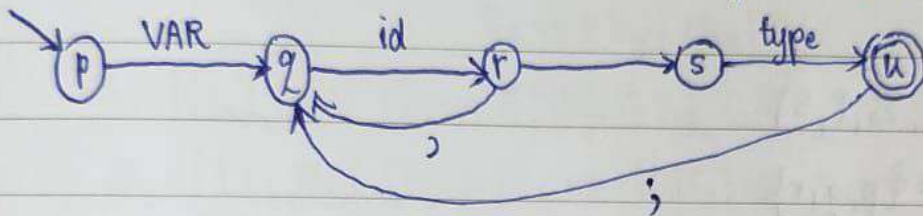
Seminar 5

example 1.

$\Sigma = \{"VAR", "type", "int", " ", ":", "(", ")", ";", "\n"\}$

VAR

b, a: integer; c, d: char;



FA $\Leftrightarrow$ RG $\Leftrightarrow$ RE

1) FA $\Leftrightarrow$ RG

example 2. $G = \{S, A\}, \{a, b\}, P, S\}$

$P: S \rightarrow aA$

$A \rightarrow aA | bA | a | b$

?FA $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{S, A, K\}$

$q_0 = S$

$\Sigma = \{a, b\}$

$F = \{K\}$

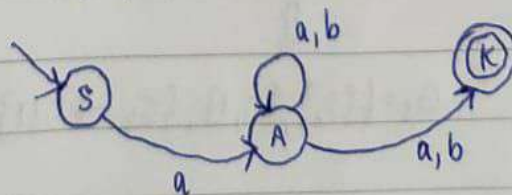
$\delta(S, a) \ni A$

$\delta(A, b) \ni K$

$\delta(A, a) \ni A$

$\delta(A, b) \ni A$

$\delta(A, a) \ni K$



example 3. $M = (Q, \Sigma, \delta, q_0, F)$

$$\Sigma = \{p, q, r\}$$

$$q_0 = p$$

$$F = \{r\}$$

$$\Sigma = \{0, 1\}$$

? RLG

δ	0	1
p	q	p
q	r	p
r	r	r

$$G = (Q, \Sigma, P, S)$$

$$N = Q = \{p, q, r\}$$

$$S = q_0 = p$$

$$P: p \rightarrow 0q \mid 1p \mid \epsilon$$

$$q \rightarrow 0r \mid 0 \mid 1p$$

$$r \rightarrow 0r \mid 0 \mid 1r \mid 1$$

2) $RG \Leftrightarrow RE$

example 4. $0(0+1)^*1$

? $\Leftrightarrow$ RG

01, 001, 011, 0001, 0101, ...

$$0: G_1 = (\{s_1\}, \{0, 1\}, \{s_1 \rightarrow 0\}, s_1)$$

$$1: G_2 = (\{s_2\}, \{0, 1\}, \{s_2 \rightarrow 1\}, s_2)$$

$$0+1: G_3 = (\{s_1, s_2, s_3\}, \{0, 1\}, \{s_1 \rightarrow 0, s_2 \rightarrow 1, s_3 \rightarrow 0 \mid 1\})$$

$\Downarrow$

$$G_3 = (\{s_3\}, \{0, 1\}, \{s_3 \rightarrow 0 \mid 1\}, s_3)$$

example 4. $(0+1)^*$

$0(0+1)^*$

$0(0+1)^*1$

example 1.

P: S -

A

B

? $\Leftrightarrow$ RE

$$\begin{cases} S = aA \\ A = aA + B \\ B = bB \end{cases}$$

$$A = aA$$

$$S = aA$$

$$S = a^+1$$

seminar 6

example 4. (continuation)

$$(0+1)^*: G_4 = (\{S_3, S_4\}, \{0, 1\}, \{S_4 \rightarrow \epsilon | 0 | 1, \\ S_3 \rightarrow 0S_4 | 1S_4 | 0 | 1\}$$

$$G'_4 = (\{S_4\}, \{0, 1\}, \{S_4 \rightarrow \epsilon | 0S_4 | 1S_4\}, S_4)$$

$$0(0+1)^*: G_5 = (\{S_3, S_1\}, \{0, 1\}, \{S_3 \rightarrow \epsilon | 0S_1 | 1S_1, S_1 \rightarrow 0S_3\}, S_1) \text{! not reg}$$

$$0(0+1)^*1: G_6 = (\{S_1, S_2, S_3\}, \{0, 1\}, \{S_2 \rightarrow 1, S_3 \rightarrow 0S_3 | 1S_3, S_2 \rightarrow 0S_3, \\ S_3 \rightarrow S_2\}, S_1) \text{! not reg}$$

$$G'_6 = (\{S_1, S_3\}, \{0, 1\}, \{S_1 \rightarrow 0S_3, S_3 \rightarrow 0S_3 | 1S_3 | 1\}, S_1)$$

example 1. $G = (\{S, A, B\}, \{0, 1\}, P, S)$

$$P: S \rightarrow aA$$

$$A \rightarrow aA | bB | b$$

$$B \rightarrow bB | b$$

$$X = aX + b$$

$$X = a^*b$$

$$aa^*b + b$$

$$= a^*b + b$$

$$= (a^* + \epsilon)b$$

$$= a^*b$$

? ($\Rightarrow$) RE

$$\begin{cases} S = aA \\ A = aA + bB + b \\ B = bB + b \end{cases} \Leftrightarrow \begin{cases} B = b^*b = b^+ \\ A = aA + B \\ S = aA \end{cases}$$

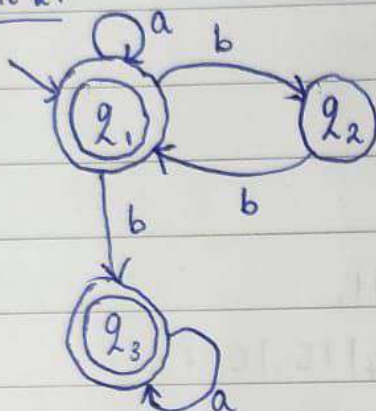
$$A = aA + B = aA + b^+ = a^*b^+$$

$$S = aa^*b^+$$

$$S = a^+b^+$$

3) FA $\Leftrightarrow$ RE

example 2.



$$X = Xa + b$$

$$X = ba^*$$

? $\Leftrightarrow$ RE

$$\begin{cases} q_1 = \epsilon + q_1 a + q_2 b \\ q_2 = q_1 b \\ q_3 = q_1 b + q_3 a \end{cases} \quad \Leftrightarrow \quad \begin{cases} q_2 = q_1 b \\ q_1 = \epsilon + q_1 a + q_1 b b \\ q_3 = q_1 a + q_3 b \end{cases}$$

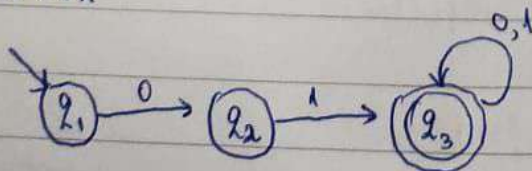
$$q_1 = \epsilon + q_1 (a + bb) \\ = (a + bb)^*$$

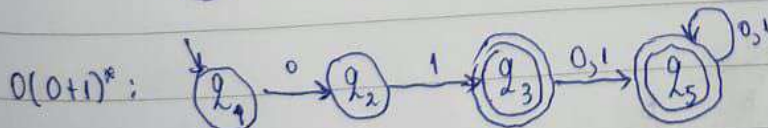
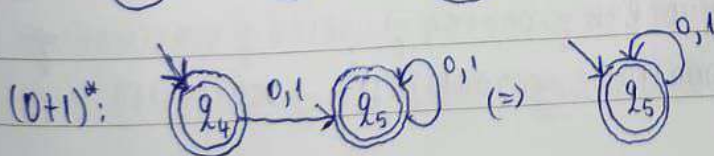
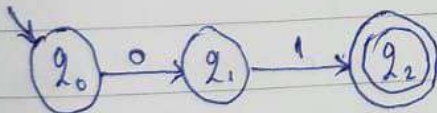
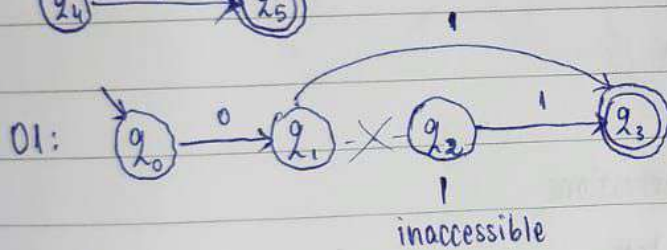
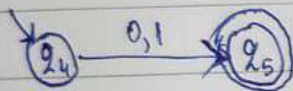
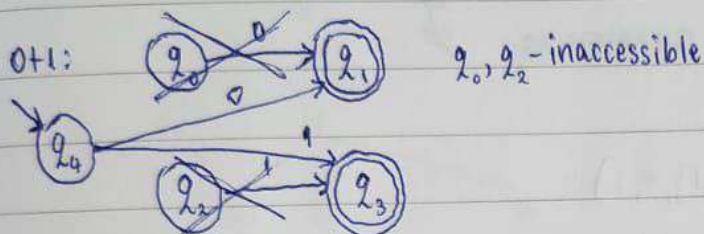
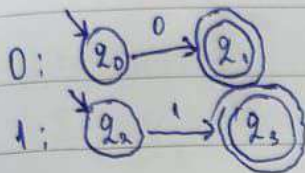
$$q_3 = (a + bb)^* b + q_1 a = (a + bb)^* b a^*$$

$$q_1 + q_3 = (a + bb)^* + (a + bb)^* b a^* \\ = (a + bb)^* (\epsilon + b a^*)$$

example 3. $01(0+1)^*1^*$ (RE)

? $\Leftrightarrow$ FA





seminar 7

CONTEXT FREE GRAMMARS

ex. 1. $G = (\{S, A, B\}, \{0, 1\}, P, S)$

$$P: S \rightarrow \overset{1}{0}B\overset{2}{1}A$$

$$A \rightarrow \overset{3}{0}\overset{4}{1}B\overset{5}{1}A$$

$$B \rightarrow \overset{6}{1}\overset{7}{1}S\overset{8}{0}BB$$

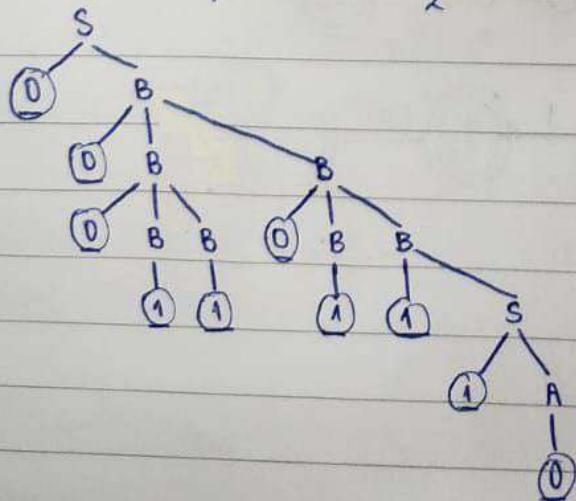
$w = 0001101110$

? leftmost / rightmost derivations

for w + corresp. parse tree

1) leftmost

$$\begin{aligned} S &\Rightarrow_1 0B \Rightarrow_8 00BB \Rightarrow_9 000BBB \Rightarrow_6 0001BB \Rightarrow_6 00011B \Rightarrow_8 000110BB \Rightarrow_6 \\ &0001101B \Rightarrow_7 00011011S \Rightarrow_2 000110111A \Rightarrow_3 0001101110 \end{aligned}$$



2) rightmost

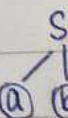
$$S \Rightarrow_1 0B \Rightarrow_8 00BB \Rightarrow_9 000BBB \Rightarrow_6 0001BB \Rightarrow_6 00011B \Rightarrow_8 000110BB \Rightarrow_6 0001101B \Rightarrow_7 00011011S \Rightarrow_2 000110111A \Rightarrow_3 0001101110$$

ex. 2 prove G

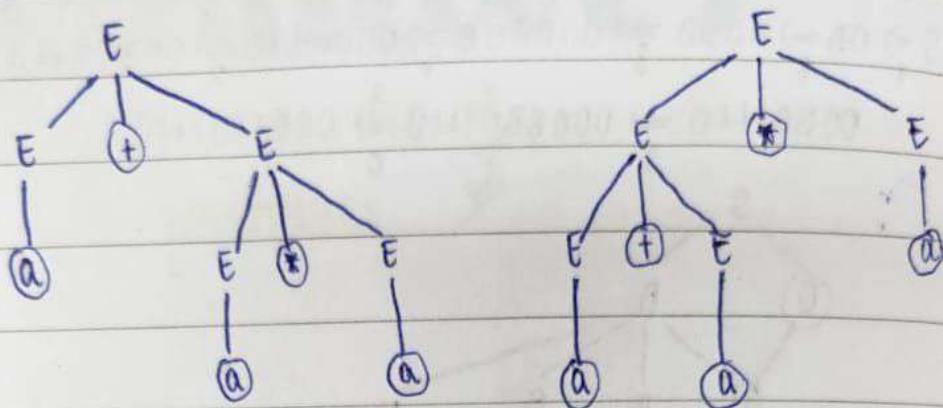
a) $G = (\{S, B\})$

b) $G = (\{E\})$

a) $w = abc$



b) $w = a + a * a$



PARSER

$(s, i, \vec{\alpha}, \leftarrow \beta)$

i.c. $(q, 1, \epsilon, S)$ (initial configuration)

f.c. $(1, n+1, \alpha, \epsilon)$ (final configuration)

RECURSIVE DESCENDENT PARSER (RDP)

$G = (\{S\}, \{a, b, c\}, P, S)$

$P: S \rightarrow aSbS \mid aS \mid c$

$w = aacbc$

? $w \in L(G)$ by RDP

$(q, 1, \epsilon, S) \xrightarrow{\text{expand}} (q, 1, S_1, aSbS) \xrightarrow{\text{advance}} (q, 2, S_1, a, SbS)$

! EXPAND only when the head of the input is a nonterminal

! ADVANCE only when the head terminal

$(q, 2, S_1, aS_1, aSbSbS) \xrightarrow{\text{advance}} (q, 3, S_1, aS_1, a, SbSbS)$

$$\vdash_{\text{expand}} (q, 3, s, a s_1 a s_1, \underline{a s b s b s b s}) \vdash_{\substack{\text{momentary} \\ \text{insuccess}}} (b, 3, s, a s_1 a s_1, \dots)$$

$$\vdash_{\substack{\text{another} \\ \text{try 1}}} (q, 3, s, a s_1 a s_2, a s b s b s) \vdash_{M_i} (b, \dots) \vdash_{AT_1} (q, 3, s, a s_1 a s_3, a b s b s)$$

$$\vdash_{\text{adv.}} (q, 4, s, a s_1 a s_3 c, b s b s) \vdash_{\text{adv.}} (q, 5, s, a s_1 a s_3 c b, s b s)$$

$$\vdash_{\text{exp.}} (q, 5, s, a s_1 a s_3 c b s_1, a s b s b s) \vdash_{M_i} (b, \dots) \vdash_{AT_1}$$

$$\vdash_{AT_1} (q, 5, s, a s_1 a s_3 c b s_2, \underline{a s b s}) \vdash_{M_i} (b, \dots) \vdash_{AT_1}$$

$$\vdash_{AT_1} (q, 5, s, a s_1 a s_3 c b s_3, c b s) \vdash_{\text{adv.}} (q, 6, s, a s_1 a s_3 c b s_3 c, b s)$$

$$\vdash_{M_i} (b, \dots) \vdash_{\text{back}} (b, 5, s, a s_1 a s_3 c b s_3, c b s) \vdash_{AT_2}$$

$$\vdash_{AT_2} (b, 5, s, a s_1 a s_3 c b, s b s) \vdash_{\text{back}}^2 (b, 3, s, a s_1 a s_3, c b s b s)$$

$$\vdash_{AT_2} (b, 3, s, a s_1 a, s b s b s) \vdash_{\text{back}} (b, 2, s, a s_1, a s b s b s) \vdash_{AT_1}$$

$$\vdash_{AT_1} (q, 2, s, a s_2, a s b s) \vdash_{\dots}$$

seminar 8

$$(q, 1, \epsilon, S) \xrightarrow{\text{exp.}} \dots \xrightarrow{\text{AT1}} (q, 2, S, S, aS_2, \underline{aSbS}) \xrightarrow{\text{adv.}} (q, 3, S, aS_2, a, SbS)$$

$$\xrightarrow{\text{exp.}} (q, 3, S, aS_2, aS_1, aSbSS) \xrightarrow{\text{M1}} (b, \dots) \xrightarrow{\text{AT1}}$$

$$\xrightarrow{\text{AT1}} (q, 3, S, aS_2, aS_2, aSbS) \xrightarrow{\text{M1}} (b, \dots) \xrightarrow{\text{AT1}} (q, 3, S, aS_2, aS_3, \text{cb})$$

$$\xrightarrow{\text{adv.}} (q, 4, S, aS_2, aS_3, \text{cb}, S) \xrightarrow{\text{adv.}} (q, 5, S, aS_2, aS_3, \text{cb}, S) \xrightarrow{\text{exp.}}$$

$$\xrightarrow{\text{exp.}} (q, 5, S, aS_2, aS_3, \text{cb}, S_1, aSbS) \xrightarrow{\text{M1, AT1, M1, AT1}}^4 (q, 5, S, aS_2, aS_3, \text{cb}, S)$$

$$\xrightarrow{\text{adv.}} (q, 6, S, aS_2, aS_3, \text{cb}, S_3, \text{c}, \epsilon) \xrightarrow{\text{succ.}} (f, 6, S, aS_2, aS_3, \text{cb}, S_3, \text{c}, \epsilon)$$

LL(1)

$$G = (\{S, A, B, C, D\}, \{a, +, *, (,)\}, P, S)$$

$$P: (1) S \rightarrow BA$$

$$(2) A \rightarrow +BA$$

$$(3) A \rightarrow \epsilon$$

$$(4) B \rightarrow DC$$

$$W = a * (a + a)$$

$$(5) C \rightarrow *DC$$

$$(6) C \rightarrow \epsilon$$

$$(7) D \rightarrow (S)$$

$$(8) D \rightarrow a$$

I FIRST +

FIRST(X)

DC

{(, a}

(
(
a
a

	F ₀
S	∅
A	+, ε
B	∅
C	*, ε
D	(, a

	L ₀
S	ε
A	∅
B	∅
C	∅
D	∅

I FIRST + FOLLOW

$$\text{FIRST}(X_1 X_2 \dots X_n) = \text{FIRST}(X_1) \oplus \text{FIRST}(X_2) \oplus \dots \oplus \text{FIRST}(X_n)$$

$\alpha \in V^*$

DC

{(, a} {*, ε}

{*, ε} {(, a}

*, (, a

(
(
a
a

	FIRST	FOLLOW
S	(, a	ε,)
A	+, ε	ε,)
B	(, a	+, ε,)
C	*, ε	+, ε,)
D	(, a	*, +, ε,)

	F ₀	F ₁	F ₂	F ₃ = F ₂ = FIRST
S	∅	∅	(, a	(, a
A	+, ε	+, ε	+, ε	+, ε
B	∅	(, a	(, a	(, a
C	*, ε	*, ε	*, ε	*, ε
D	(, a	(, a	(, a	(, a

	L ₀	L ₁	L ₂	L ₃	L ₄ = L ₃ = FOLLOW
S	ε	ε,)	ε,)	ε,)	ε,)
A	∅	ε	ε,)	ε,)	ε,)
B	∅	+, ε	+, ε,)	+, ε,)	+, ε,)
C	∅	∅	+, ε	+, ε,)	+, ε,)
D	∅	*	*, +, ε	*, +, ε	*, +, ε,)

II LL(1) Parse Table

	a	+	*	(	)	\$
S	BA, 1			BA, 1		
A		+BA, 2			E, 3	E, 3
B	DC, 4			DC, 4		
C		E, 6	*DC, 5		E, 6	E, 6
D	a, 8			(S), 7		
a	pop					
+		pop				
*			pop			
(				pop		
)					pop	
\$						accept

III ANALYSIS

$(a*(a+a)\$, S\$, \epsilon) \vdash (a*(a+a)\$, BA\$, 1) \vdash (a*(a+a)\$, DC\$, 4)$
 $\vdash (a*(a+a), aCA\$, 148) \xrightarrow{\text{pop}} (* (a+a), CA\$, 148)$

LL(1) ANALYSIS

$(*(a+a)\$, ($

$\vdash ($

$\vdash ($

$\vdash ($

$\vdash ($

$\vdash \text{pop}$

$\vdash ($

$\vdash 2$

$\vdash 2$

how to transform

(1) $A \rightarrow \alpha$

(2) $A \rightarrow \alpha$

if stmt $\rightarrow$

if stmt $\rightarrow$

$B \rightarrow E$

seminar 9

LL(1) III ANALYSIS (continuation)

$$\begin{aligned}
 & (* (a+a) \$, CA \$, 148) \vdash (* (a+a) \$, * DCA \$, 1485) \vdash_{\text{pop}} ((a+a) \$, DCA \$, 1485) \\
 & \vdash ((a+a) \$, (S) CA, 14857) \vdash_{\text{pop}} ((a+a) \$, S) CA \$, 14857) \vdash \\
 & \vdash ((a+a) \$, BA) CA \$, 148571) \vdash ((a+a) \$, DCA) CA \$, 1485714) \\
 & \vdash ((a+a) \$, aCA) CA \$, 14857148) \vdash_{\text{pop}} ((a) \$, CA) CA \$, 14857148) \\
 & \vdash ((a) \$, A) CA \$, 148571486) \vdash ((a) \$, +BA) CA \$, 1485714862) \\
 & \vdash_{\text{pop}} ((a) \$, BA) CA \$, 1485714862) \vdash ((a) \$, DCA) CA \$, 14857148624) \\
 & \vdash ((a) \$, aCA) CA \$, 148571486248) \vdash_{\text{pop}} (() \$, CA) CA \$, 14857148624) \\
 & \vdash^2 (() \$,) CA \$, 1485714862463) \vdash_{\text{pop}} ((\$, CA \$, 1485714862463) \\
 & \vdash^2 ((\$, \$, 148571486246363) \vdash_{\text{accept}}
 \end{aligned}$$

how to transform if not in LL(1)?

- (1) $A \rightarrow \alpha \beta \quad A \rightarrow \alpha B$
- (2) $A \rightarrow \alpha \gamma \Rightarrow B \rightarrow \beta \mid \gamma$

if stmt $\rightarrow$ if cond then stmt

if cond then stmt else stmt

if stmt $\rightarrow$ if cond then stmt B

$B \rightarrow \epsilon \mid \text{else stmt}$

LR(0)

example 1. $G = (\{S, A\}, \{a, b, c\}, P, S)$
 $S' \rightarrow S$

P: (1) $S \rightarrow aA$

(2) $A \rightarrow bA$

(3) $A \rightarrow c$

$w = abbc \in L(G)$

I Compute the canonical collection of states

$$s_0 = \text{closure}(\{[S' \rightarrow \cdot S]\}) = \{[S' \rightarrow \cdot S], [S \rightarrow \cdot aA]\} \quad \left| \begin{array}{l} [A \rightarrow \alpha \cdot \beta] - \text{LR(0) item} \\ \text{Kernel} \end{array} \right.$$

$$s_1 = \text{goto}(s_0, S) = \text{closure}(\{[S' \rightarrow S \cdot]\}) = \{[S' \rightarrow S \cdot]\}$$

$$\text{goto}(s_0, A) = \text{closure}(\emptyset) = \emptyset$$

$$s_2 = \text{goto}(s_0, a) = \text{closure}(\{[S \rightarrow a \cdot A]\}) = \{[S \rightarrow a \cdot A], [A \rightarrow \cdot bA], [A \rightarrow \cdot c]\}$$

$$\text{goto}(s_0, b) = \text{closure}(\emptyset) = \emptyset$$

$$\text{goto}(s_0, c) = \emptyset$$

$$s_3 = \text{goto}(s_2, A) = \text{closure}(\{[S \rightarrow aA \cdot]\}) = \{[S \rightarrow aA \cdot]\}$$

$$s_4 = \text{goto}(s_2, b) = \text{closure}(\{[A \rightarrow b \cdot A]\}) = \{[A \rightarrow b \cdot A], [A \rightarrow \cdot bA], [A \rightarrow \cdot c]\}$$

$$s_5 = \text{goto}(s_2, c) = \text{closure}(\{[A \rightarrow c \cdot]\}) = \{[A \rightarrow c \cdot]\}$$

$$s_6 = \text{goto}(s_4, A) = \text{closure}(\{[A \rightarrow bA \cdot]\}) = \{[A \rightarrow bA \cdot]\}$$

$$\text{goto}(s_4, b) = \text{closure}(\{[A \rightarrow b \cdot A]\}) = s_4$$

$$\text{goto}(s_4, c) = \text{closure}(\{[A \rightarrow c \cdot]\}) = s_5$$

$$C = \{s_0, s_1, \dots, s_6\}$$

II LR(0) pars

	A
0	SHIFT
1	ACCEPT
2	SHIFT
3	REDUCE
4	SHIFT
5	REDUCE
6	REDUCE

III Analysis

WORK STA

\$ 0

\$ 0 a 2

\$ 0 a 2 b 4

\$ 0 a 2 b 4 b

\$ 0 a 2 b 4 b 4

\$ 0 a 2 b 4 b 4

\$ 0 a 2 b 4 A 6

\$ 0 a 2 A 3

\$ 0 S 1

accepted.

II LR(0) parsing table

	ACTION	GOTO				
		S	A	a	b	c
0	SHIFT	1		2		
1	ACCEPT					
2	SHIFT		3		4	5
3	REDUCE (1)					
4	SHIFT		6		4	5
5	REDUCE (3)					
6	REDUCE (2)					

III Analysis

WORK STACK	INPUT STACK	OUTPUT BAND
\$0	abbc\$	ϵ
\$0a2	bbc\$	ϵ
\$0a2b4	bc\$	ϵ
\$0a2b4b4	c\$	ϵ
\$0a2b4b4c5	\$	ϵ
\$0a2b4b4AG	\$	3
\$0a2b4AG	\$	13
\$0a2A3	\$	223
\$0S1	\$	1223

accepted.

seminar 10

SLR

$$G = (\{E, T\}, \{id, const, +, (,)\}, P, S)$$

$$P \vdash (1) \overset{S' \rightarrow E}{E} \rightarrow T$$

$$(2) E \rightarrow E + T$$

$$(3) T \rightarrow (E)$$

$$(4) T \rightarrow id$$

$$(5) T \rightarrow const$$

$$w = id + const \in L(G)$$

I Canonical collection

$$s_0 = \text{closure}(\{[S' \rightarrow \cdot E]\}) = \{[S' \rightarrow \cdot E], [E \rightarrow \cdot T], [E \rightarrow \cdot E + T], [T \rightarrow \cdot (E)], [T \rightarrow \cdot id], [T \rightarrow \cdot const]\}$$

$$s_1 = \text{goto}(s_0, E) = \text{closure}(\{[S' \rightarrow E \cdot], [E \rightarrow E \cdot + T]\}) = \{[S' \rightarrow E \cdot], [E \rightarrow E \cdot + T]\}$$

$$s_2 = \text{goto}(s_0, T) = \text{closure}(\{[E \rightarrow T \cdot]\}) = \{[E \rightarrow T \cdot]\}$$

$$s_3 = \text{goto}(s_0, () = \text{closure}(\{[T \rightarrow (\cdot E)]\}) = \{[T \rightarrow (\cdot E)], [E \rightarrow \cdot E + T], [E \rightarrow \cdot T], [T \rightarrow (\cdot (E)], [T \rightarrow \cdot id], [T \rightarrow \cdot const]\}$$

$$s_4 = \text{goto}(s_0, id) = \text{closure}(\{[T \rightarrow id \cdot]\}) = \{[T \rightarrow id \cdot]\}$$

$$s_5 = \text{goto}(s_0, const) = \text{closure}(\{[T \rightarrow const \cdot]\}) = \{[T \rightarrow const \cdot]\}$$

$$s_6 = \text{goto}(s_1, +) = \text{closure}(\{[E \rightarrow E + \cdot T]\}) = \{[E \rightarrow E + \cdot T], [T \rightarrow \cdot (E)], [T \rightarrow \cdot id], [T \rightarrow \cdot const]\}$$

$$s_7 = \text{goto}(s_3, E) = \text{closure}(\{[T \rightarrow (E \cdot)], [E \rightarrow E \cdot + T]\}) = \{[T \rightarrow (E \cdot)], [E \rightarrow E \cdot + T]\}$$

$$\text{goto}(s_3, T)$$

$$\text{goto}(s_3, ($$

$$\text{goto}(s_3, id$$

$$\text{goto}(s_3, const$$

$$s_8 = \text{goto}(s_6, T$$

$$\text{goto}(s_6, ($$

$$\text{goto}(s_6, id$$

$$\text{goto}(s_6, const$$

$$s_9 = \text{goto}(s_7, T$$

$$\text{goto}(s_7, ($$

II SLR Parsing

$$\text{FOLLOW}(E) =$$

$$\text{FOLLOW}(T) =$$

	id	
0	SHIFT 4	
1		
2		
3	SHIFT 4	
4		
5		
6	SHIFT 5	
7		
8		
9		

$$\text{goto}(s_3, T) = \text{closure}(\{[E \rightarrow T.] \}) = s_1$$

$$\text{goto}(s_3, () = \text{closure}(\{[T \rightarrow (.E)] \}) = s_3$$

$$\text{goto}(s_3, \text{const}) = s_5$$

$$\text{goto}(s_3, \text{id}) = s_4$$

$$s_2 = \text{goto}(s_6, T) = \text{closure}(\{[E \rightarrow E+T.] \}) = \{[E \rightarrow E+T.] \}$$

$$\text{goto}(s_6, () = s_3$$

$$\text{goto}(s_6, \text{id}) = s_4$$

$$\text{goto}(s_6, \text{const}) = s_5$$

$$s_9 = \text{goto}(s_7,) = \{[E \rightarrow (T).] \}$$

$$\text{goto}(s_7, +) = s_6$$

II SLR Parsing Table

$$\text{FOLLOW}(E) = \{E, +,)\}$$

$$\text{FOLLOW}(T) = \{E, +,)\}$$

	ACTION						GOTO	
	id	const	+	(	)	\$	E	T
0	SHIFT 4	SHIFT 5		SHIFT 3			1	2
1			SHIFT 6			ACC.		
2			RED. 1	RED. 1		RED. 1		
3	SHIFT 4	SHIFT 5		SHIFT 3			7	2
4			RED. 4		RED. 4	RED. 4		
5			RED. 5		RED. 5	RED. 5		
6	SHIFT 5	SHIFT 4		SHIFT 3				8
7			SHIFT 6			SHIFT 3		
8			RED. 2		RED. 2	RED. 2		
9			RED. 3		RED. 3	RED. 3		

III Analysis

WORK STACK	INPUT STACK	OUTPUT BAND
\$0	id+const \$	E
\$0 id 4	+ const \$	E
\$0 T 2	+ const \$	4
\$0 E 1	+ const \$	14
\$0 E 1 + 6	const \$	14
\$0 E 1 + 6 const 5	\$	14
\$0 E 1 + 6 T 8	\$	514
\$0 E 1	\$	2514
accept		

Seminar 11

LR(1) parser

$$G = (\{S, A\}, \{a, b\}, P, S)$$

$$S' \rightarrow S$$

$$P: (1) S \rightarrow AA$$

$$(2) A \rightarrow aA$$

$$(3) A \rightarrow b$$

$$w = abab \in L(G)$$

$$\text{LR(1) item: } [A \rightarrow \alpha, \beta, a] \\ \text{where } \beta \in \Sigma \cup \{\$ \}$$

first of ternal - a set of

$$\text{FIRST}(A) = \{a, b\}$$

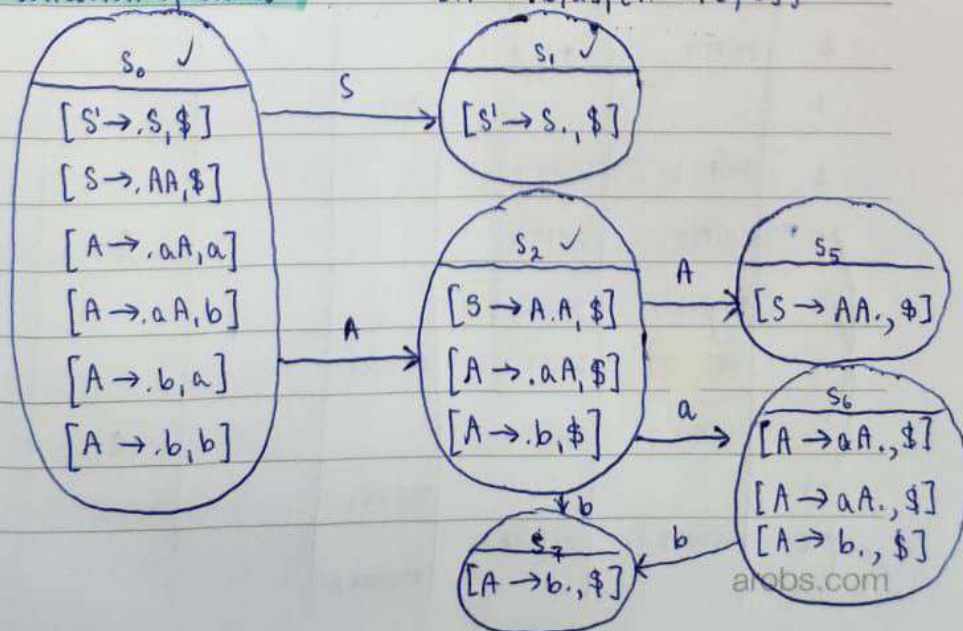
$$\text{FIRST}(S) = \{a, b\}$$

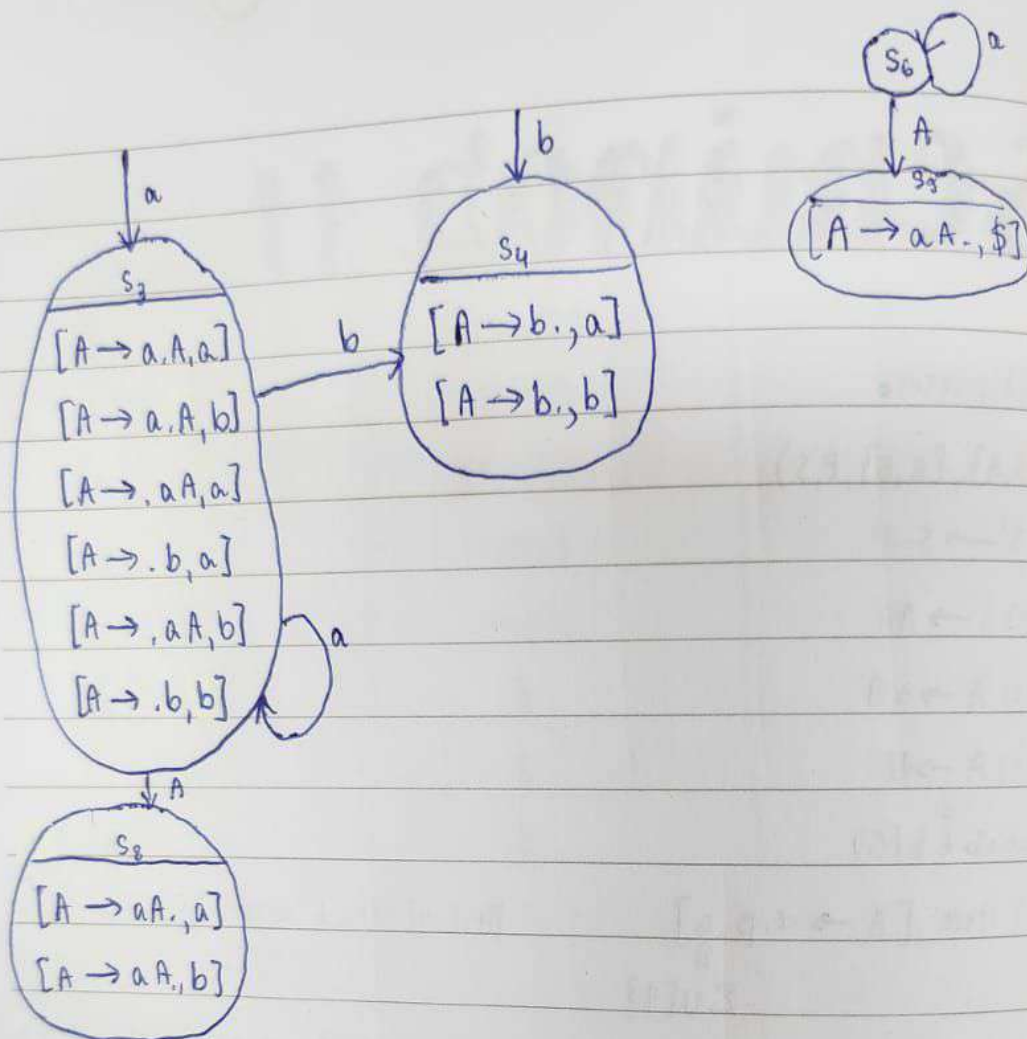
$$s_0 = \text{closure}(\{[S' \rightarrow \cdot S, \$]\}) = \{[S' \rightarrow \cdot S, \$],$$

$$[S \rightarrow \cdot AA, \$], [A \rightarrow \cdot aA, a], [A \rightarrow \cdot aA, b],$$

$$[A \rightarrow \cdot b, a], [A \rightarrow \cdot b, b]\}$$

I Canonical collection of states





II Parsing table

	ACTION			GOTO	
	a	b	\$	S	A
0	SHIFT 3	SHIFT 4		1	2
1			ACC.		
2	SHIFT 6	SHIFT 7			
3	SHIFT 3	SHIFT 4			5
4	REDUCE 3	REDUCE 3			8
5			REDUCE 1		
6	SHIFT 6	SHIFT 7			
7			REDUCE 1		
8	REDUCE 2	REDUCE 2	REDUCE 3		9
9			REDUCE 2		

III Analysis

WORK

\$0

\$0a3

\$0a3b4

\$0a3A9

\$0A2

\$0A2a6

\$0A2a6

\$0A2a6

\$0A2A9

\$0S1

acc.

LALR

I Canonical

$$S_8 + S_9 =$$

$$S_{89} = \{$$

$$S_4 + S_7 =$$

$$S_{47} = \{$$

$$S_3 + S_6 =$$

$$S_{36} = \{$$

$$B = \{S_0$$

III Analysis

WORK STACK	INPUT STACK	OUTPUT BAND
\$0	abab\$	ϵ
\$0a3	bab\$	ϵ
\$0a3b4	ab\$	ϵ
\$0a3A8	ab\$	3
\$0A2	ab\$	23
\$0A2a6	b\$	23
\$0A2a6b7	\$	23
\$0A2a6A9	\$	323
\$0A2A5	\$	2323
\$0S1	\$	12323
acc.		

LALR

I Canonical collection of states

$$S_8 + S_9 \rightarrow S_{89}$$

$$S_{89} = \{[A \rightarrow aA, alb\$]\}$$

$$S_4 + S_7 = S_{47}$$

$$S_{47} = \{[A \rightarrow b, alb\$]\}$$

$$S_3 + S_6 = S_{36}$$

$$S_{36} = \{[A \rightarrow aA, alb\$],$$

$$[A \rightarrow aA, alb\$],$$

$$[A \rightarrow b, alb\$]\}$$

$$C = \{S_0, S_1, S_2, S_{36}, S_{47}, S_{89}, S_5\}$$

II Parsing table

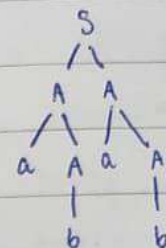
	ACTION		GOTO		
	a	b	\$	S	A
0	SHIFT 36	SHIFT 47		1	2
1			ACC.		
2	SHIFT 36	SHIFT 47			5
36	SHIFT 36	SHIFT 47			89
47	REDUCE 3	REDUCE 3	REDUCE 3		
5			REDUCE 1		
89	REDUCE 2	REDUCE 2			

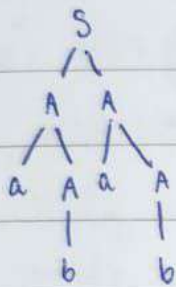
III Analysis

WORK STACK	INPUT STACK	OUTPUT BAND
\$0	abab\$	ε
\$0a36	bab\$	ε
\$0a36b47	ab\$	ε
\$0a36A89	ab\$	3
\$0A2	ab\$	23
\$0A2a36	b\$	23
\$0A2a36b47	\$	23
\$0A2a36A89	\$	23
\$0A2A5	\$	323
\$0S1	\$	2323
acc.	\$	12323

R₁: 12323

R₂: S ⇒ AA ⇒ AaA ⇒ Aab ⇒ aAab ⇒ abab





index	symbol	F	S
0	S	-1	-1
1	A	0	2
2	A	0	-1
3	a	1	4
4	A	1	-1

seminar 12

PDA

$$M = (Q, \Sigma, \overset{\text{stack memory alphabet}}{\Gamma}, \delta, q_0, z_0, F)$$

↓ initial stack symbol

$$FA: \delta: Q \times \Sigma \rightarrow \mathcal{P}(Q)$$

$$PDA: \delta: Q \times \Sigma \cup \{\epsilon\} \times \Gamma \rightarrow \mathcal{P}(Q \times \Gamma^*)$$

$$L_f(M) = \{w \in \Sigma^* \mid (q_0, w, z_0) \xrightarrow{*} (q_f, \Sigma, \gamma), \gamma \in \Gamma, q_f \in F\} \rightarrow \text{by the final state criterion}$$

$$L_\epsilon(M) = \{w \in \Sigma^* \mid (q_0, w, z_0) \xrightarrow{*} (q, \epsilon, \epsilon), q \in Q\} \rightarrow \text{by the empty stack criterion}$$

reverse

example 1. ?PDA $L_1 = \{w w^R \mid w \in \{a, b\}^+\}$, $\overset{R}{a}\overset{R}{a}, \overset{R}{b}\overset{R}{b}, \overset{R}{a}\overset{R}{b}\overset{R}{b}\overset{R}{a}, \overset{R}{a}\overset{R}{b}\overset{R}{b}\overset{R}{b}\overset{R}{b}\overset{R}{a}, \dots$

$$M = (\{q_0, q_1\}, \{a, b\}, \{Z_0, A, B\}, \delta, q_0, Z_0, \epsilon)$$

$$\delta(q_0, a, Z_0) = (q_1, AZ_0)$$

$$\delta(q_0, b, Z_0) = (q_1, BZ_0)$$

$$\delta(q_0, a, A) = (q_1, AA)$$

$$\delta(q_0, a, B) = (q_1, AB)$$

$$\delta(q_0, b, B) = (q_1, BB)$$

$$\delta(q_0, b, A) = (q_1, BA)$$

$$\delta(q_0, a, A) = (q_1, \epsilon)$$

$$\delta(q_0, b, B) = (q_1, \epsilon)$$

$$\delta(q_1, a, A) = (q_1, \epsilon)$$

$$\delta(q_1, b, B) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, Z_0) = (q_1, \epsilon)$$

2. $L_2 = \{$

3. $L_3 = \{$

4. $L_4 = \{$

2.

3.

$b, A \rightarrow \epsilon$

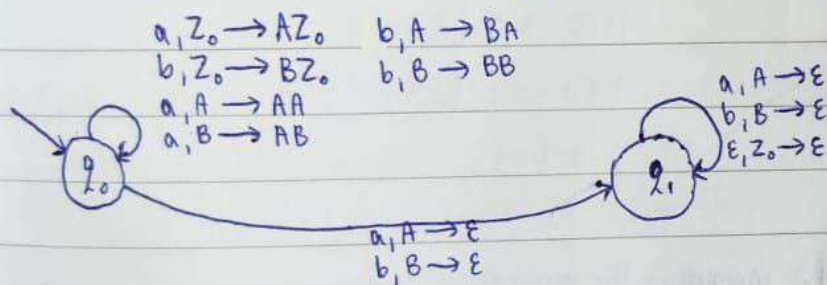
$b, A \rightarrow \epsilon$

$ww^R = abbaabba$

$(q_0, abbaabba, Z_0) \vdash (q_0, bbaabba, AZ_0) \vdash (q_0, baabba, BAZ_0) \vdash$
 $\vdash (q_0, aabba, BB AZ_0) \vdash (q_0, abba, ABB AZ_0) \vdash (q_1, bba, BB AZ_0) \vdash$
 $\vdash (q_1, ba, BAZ_0) \vdash (q_1, a, AZ_0) \vdash (q_1, \epsilon, Z_0) \vdash (q_1, \epsilon, \epsilon)$

$ww^R = abbabaab$

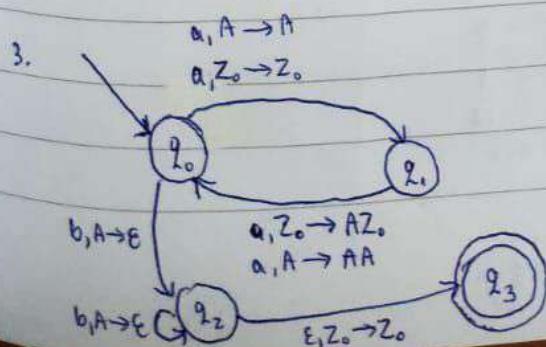
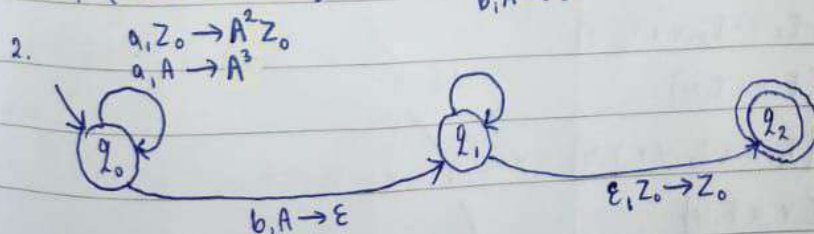
$(q_0, abbabaab, Z_0) \vdash \dots (q_0, baab, ABB AZ_0) \vdash (q_0, aab, BABBAZ_0) \vdash$
 $\vdash (q_0, ab, ABABBAZ_0) \vdash (q_0, b, AABABBAZ_0) \vdash (q_0, \epsilon, BAABABBAZ_0)$
 $\vdash (q_1, b, BABBAZ_0) \vdash (q_1, \epsilon, A \dots)$



2. $L_2 = \{a^n b^{2n} \mid n \in \mathbb{N}^*\}$

3. $L_3 = \{a^{2n} b^n \mid n \in \mathbb{N}^*\}$

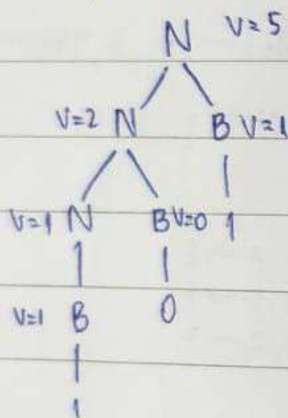
4. $L_4 = \{a^n b^{2n} \mid n \in \mathbb{N}\}$



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ATTRIBUTE GRAMMARS

CFG
(G, A, R)



$$(1) N \rightarrow NB \{N_1.v = 2 * N_2.v + B.v\}$$

$$(2) N \rightarrow B \{N.v = B.v\}$$

$$(3) B \rightarrow 0 \{B.v = 0\}$$

$$(4) B \rightarrow 1 \{B.v = 1\}$$

$$A = \{v\}$$

exercise 1. AG for computing the number of vowels in a non-empty string

exercise 2. AG for computing the value of an arithmetic expression with $+, *, (,)$.

$$G = (\{E, T, F\}, \{+, *, (,), id\}, P, E)$$

$$E \rightarrow E + T \{E.v = E_1.v + T.v\}$$

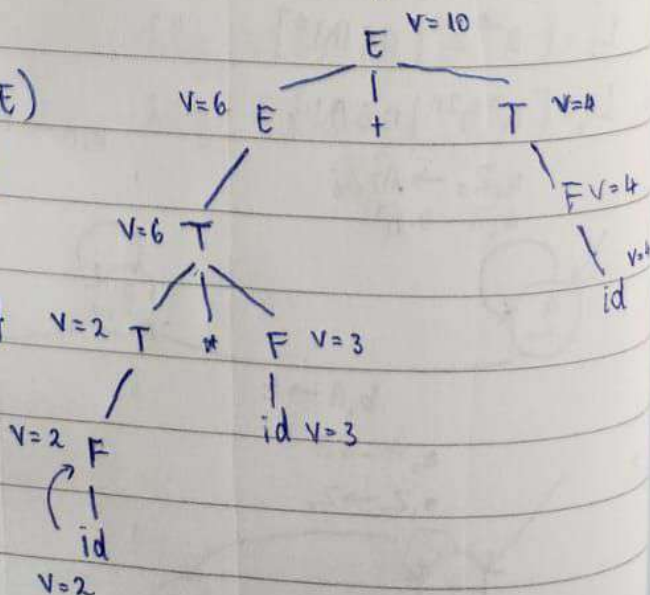
$$E \rightarrow T \{E.v = T.v\}$$

$$T \rightarrow T * F \{T.v = T_1.v * F.v\}$$

$$T \rightarrow F \{T.v = F.v\}$$

$$F \rightarrow (E) \{F.v = E.v\}$$

$$F \rightarrow id \{F.v = id.v\}$$



exercise 1.

$S \rightarrow$

$S \rightarrow$

$L \rightarrow$

$L \rightarrow$

$V = 0$

$C = b$

abc

exercise 3. A

$N \rightarrow$

$N \rightarrow$

$A \rightarrow$

$A \rightarrow$

$D \rightarrow$

$D \rightarrow$

$S \rightarrow$

$S \rightarrow$

753

exercise 1. $G = (\{S, L, C, V\}, \{a, b, c, \dots, z\}, P, s)$

$S \rightarrow L \quad \{s, n = L.n\}$

$S \rightarrow SL \quad \{S_1.n = S_2.n + L.n\}$

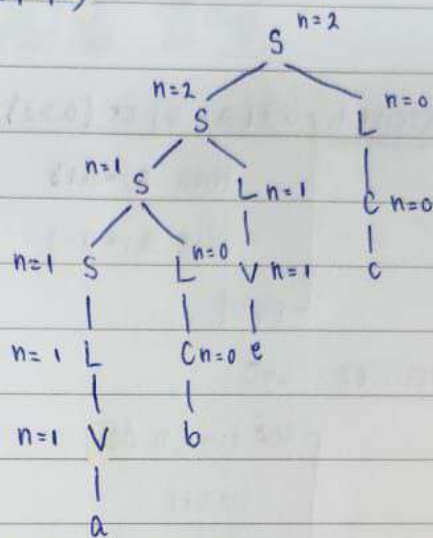
$L \rightarrow V \quad \{L.n = V.n\}$

$L \rightarrow C \quad \{L.n = C.n\}$

$V = a|e|i|l|o|u \quad \{C.n = 1\}$

$C = b|c| \dots |z \quad \{C.n = 0\}$

abec



exercise 3. AG for verifying a natural number is divisible by 3.

$N \rightarrow D \quad \{N.s = D.s; N.d = (D.s \% 3 == 0)\}$

$N \rightarrow AS \quad \{N.s = A.s + S.s; N.d = (N.s \% 3 == 0)\}$

$A \rightarrow 1 \quad \{A.s = 1\}$

$A \rightarrow 9 \quad \{A.s = 9\}$

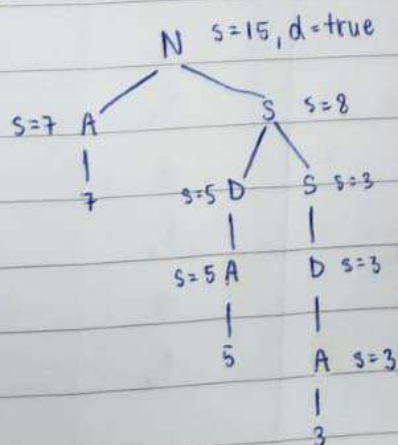
$D \rightarrow 0 \quad \{D.s = 0\}$

$D \rightarrow A \quad \{D.s = A.s\}$

$S \rightarrow D \quad \{S.s = D.s\}$

$S \rightarrow DS \quad \{S_1.s = D.s + S_2.s\}$

753



3-ADDRESS CODE

exercise 1. if $(a < b)$ or $(a > 3)$ and $(a < 10)$
 then $a := a + b$
 else $a := a - 3$
 endif

exercise 2. $s = 0$

for $i := 1, n$ do
 $s := s + i$
 endfor

exercise 1.

label	operator	arg 1	arg 2	result
1	<	a	b	t_1
2	>	a	3	t_2
3	<	a	10	t_3
4	and	t_2	t_3	t_4
5	or	t_1	t_4	t_5
6	goto	t_5		(9)
7	-	a	3	t_7
8	:=	t_7		a
9	goto			(12)
10	+	a	b	t_8
11	:=			a
12		t_8		

exercise 2. $s = 0$

label

1
2
3
4
5
6
7
8
9
10
11
12

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exercise 2. $s := 0$

```

for i := 1, n do
    b := -i;
    s := s + b;
endfor
  
```

label	operator	arg 1	arg 2	result
1	$:=$	0		s
2	$:=$	1		i
3	$>$	i	n	t_1
4	goto	t_1		(12)
5	@ (-unary)	i		t_2
6	$:=$	t_2		b
7	+	s	b	t_3
8	$:=$	t_3		s
9	+	i	1	t_4
10	$:=$	t_4		i
11	goto			3
12				